Moore Machine

Definition A Moore machine is a six-tuple $(Q, \Sigma, \Delta, \delta, \lambda, q_0)$, where

(i) Q is a finite set of states;
(ii) Σ is the input alphabet;
(iii) Δ is the output alphabet;
(iv) δ is the transition function $\Sigma \times Q$ into Q;
(v) λ is the output function mapping Q into Δ; and
(vi) q_0 is the initial state.

Mealy Machine

Definition A Mealy machine is a six-tuple $(Q, \Sigma, \Delta, \delta, \lambda, q_0)$, where all the symbols except λ have the same meaning as in the Moore machine. λ is the output function mapping $\Sigma \times Q$ into Δ.

For example, Table 3.8 describes a Moore machine. The initial state q_0 is marked with an arrow. The table defines δ and λ.

Note:-

TABLE 3.8 A Moore Machine

Present state	Next state δ		Output
	$a = 0$	$a = 1$	λ
$\rightarrow q_0$	q_3	q_1	0
q_1	q_1	q_2	1
q_2	q_2	q_3	0
q_3	q_3	q_0	0

For the input string 0111, the transition of states is given by $q_0 \rightarrow q_3 \rightarrow q_0 \rightarrow q_1 \rightarrow q_2$. The output string is 00010. For the input string Λ, the output is $\lambda(q_0) = 0$.

Transition Table 3.9 describes a Mealy machine.

TABLE 3.9 A Mealy Machine

Present state	Next state			
	$a = 0$		$a = 1$	
	state	output	state	output
$\rightarrow q_1$	q_3	0	q_2	0
q_2	q_1	1	q_4	0
q_3	q_2	1	q_1	1
q_4	q_4	1	q_3	0

Note: For the input string 0011, the transition of states is given by $q_1 \rightarrow q_3 \rightarrow q_2 \rightarrow q_4 \rightarrow q_3$, and the output string is 0100. In the case of a Mealy machine, we get an output only on the application of an input symbol. So for the input string Λ, the output is only Λ. It may be observed that in the case of a Moore machine, we get $\lambda(q_0)$ for the input string Λ.

Note:-

For a Moore machine if the input string is of length n, the output string is of length $n + 1$. The first output is $\lambda(q_0)$ for all output strings. In the case of a Mealy machine if the input string is of length n, the output string is also of the same length n.

Q. Construct the Moore machine from the following given Mealy machine

TABLE 3.10 Mealy Machine of Example 3.9

Present state	Next state			
	Input $a = 0$		Input $a = 1$	
	state	output	state	output
$\rightarrow q_1$	q_3	0	q_2	0
q_2	q_1	1	q_4	0
q_3	q_2	1	q_1	1
q_4	q_4	1	q_3	0

Solution

TABLE 3.11 State Table for Example 3.9

Present state	*Next state*			
	Input a = 0		*Input a = 1*	
	state	*output*	*state*	*output*
$\rightarrow q_1$	q_3	0	q_{20}	0
q_{20}	q_1	1	q_{40}	0
q_{21}	q_1	1	q_{40}	0
q_3	q_{21}	1	q_1	1
q_{40}	q_{41}	1	q_3	0
q_{41}	q_{41}	1	q_3	0

TABLE 3.12 Revised State Table for Example 3.9

Present state	*Next state*		*Output*
	$a = 0$	$a = 1$	
$\rightarrow q_1$	q_3	q_{20}	1
q_{20}	q_1	q_{40}	0
q_{21}	q_1	q_{40}	1
q_3	q_{21}	q_1	0
q_{40}	q_{41}	q_3	0
q_{41}	q_{41}	q_3	1